



Faculty of Computing and Technology

Department of Computing

Bachelor of Science (Hons) in Computer Science

End - Semester Examination

Year 2026, Semester 2 - Batch 3

CS 1208 – Operating Systems (Theory)

Duration: 2 Hours

Instructions to the candidates:

1. Answer **ALL** the questions.
2. Illustrate your answers with clear diagrams wherever applied.
3. The paper is marked out of 100 Marks.
4. Follow the General Guidelines given by the Department of Examination for exam Start time and closing time.

Question 01

- a) List **FOUR (04)** types of Operating Systems in the Operating Systems Landscape. (4 Marks)
- b) Identify **FOUR (04)** objectives of an Operating System. (4 Marks)
- c) In the context of Operating Systems, describe the following views. (6 Marks)
 - i. Operating System as an Extended Machine (Top – Down View)
 - ii. Operating System as a Resource Manager (Bottom – Up View)

d) In the 2nd Generation of the evolution of Operating systems, **Multiprogramming** was introduced. Define what is meant by **Multiprogramming**.

(4 Marks)

e) State **Two (02)** Future trends in Operating Systems.

(2 Marks)

[Total = 20 Marks]

Question 02

a) In the context of Operating Systems, describe what is meant by a **Process**.

(5 Marks)

b) Compare the difference between a **Program** and a **Process**.

(5 Marks)

c) Using a suitable diagram, identify the **Different States** of a Process.

(7 Marks)

d) In the context of Unix / Linux operating systems, describe the purpose of **fork()** method.

(3 Marks)

[Total = 20 Marks]

Question 03

a) Identify the **Two (02)** types of Parallelisms in the context of Operating Systems.

(2 Marks)

b) Compare the difference between **User Level Threads (ULTs)** and **Kernal Level Threads (KLTs)**.

(6 Marks)

c) Consider the following three processes.

Process	Burst Time
P1	6
P2	8
P3	4

Given that all the processes arrived at time 0, using **Shortest Job First Scheduling** to calculate the following

- i. Completion time for each process
- ii. Turnaround time for each process
- iii. Waiting time for each process
- iv. Average Turnaround time
- v. Average Waiting time

(12 Marks)

[Total = 20 Marks]

Question 04

- a) In the context of concurrency handling in operating systems, define what is meant by a **Semaphore**.
(3 Marks)
- b) List **Two (02)** Classic Synchronization Problems found in Operating System Theory.
(2 Marks)
- c) List **Two (02)** Addressing Schemes in Message Passing. *over n*
(4 Marks)
- d) State the **FOUR (04)** necessary conditions which must hold simultaneously for a Deadlock to occur.
(8 Marks)
- e) With the use of a suitable **Resource Trajectory Graph**, illustrate the following components.
 - i. State Point
 - ii. Forbidden Region
 - iii. Deadlock Point

(3 Marks)

[Total = 20 Marks]

Question 05

- a) List **three (03)** critical functions which an operating system performs to manage the memory.
(3 Marks)
- b) Identify **one (01) advantage** and a **one (01) disadvantage** of following memory partitioning mechanisms.
 - i. Fixed Partitioning

- ii. Dynamic Partitioning
- iii. Buddy System

(6 Marks)

c) Define what is meant by **Virtual Memory**.

(3 Marks)

d) Describe the following concepts in Virtual Memory

- i. Logical Address
- ii. Physical Address
- iii. Pages
- iv. Page Table

(8 Marks)

[Total = 20 Marks]

***** End of the Paper *****